11.4.4 Distribution function from Probability density function

Both the probability density function and the cumulative distribution function (or distribution function) of a continuous random variable X contain all the probabilistic information of X. The *probability distribution* of X is determined by either of them. Let us learn the method to determine the distribution function F of a continuous random variable X from the probability density function $f(x)$ of X and vice versa.

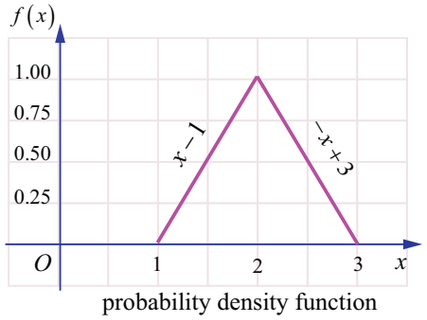
Example 11.12

If X is the random variable with probability density function $f(x)$ given by,

$$f(x)=\begin{cases} x-1, & 1\le x<2 \\ -x+3, & 2\le x<3 \\ 0 & \text{otherwise} \end{cases}$$

find (i) the distribution function $F(x)$

(ii) $P(1.5\le X\le 2.5)$



probability density function

Fig. 11.16

Solution

(i) By definition $F(x)=P(X\le x)=\int_{-\infty}^{x} f(u)\,du$

When $x<1$ $\quad F(x)=P(X\le x)=\int_{-\infty}^{x} 0\,du=0$.

When $1\le x<2$ $\quad F(x)=P(X\le x)=\int_{-\infty}^{1} 0\,du+\int_{1}^{x}(u-1)\,du$

$$=0+\left[\frac{(u-1)^2}{2}\right]_1^x=\frac{(x-1)^2}{2}$$

When $2\le x<3$ $\quad F(x)=P(X\le x)=\int_{-\infty}^{1} 0\,du+\int_{1}^{2}(u-1)\,du+\int_{2}^{x}(3-u)\,du$

$$=0+\left[\frac{(u-1)^2}{2}\right]_1^2+\left[-\frac{(3-u)^2}{2}\right]_2^x$$

$$=\frac{1^2-0}{2}+\frac{1-(3-x)^2}{2}=1-\frac{(3-x)^2}{2}$$

When $x\ge 3$, $\quad F(x)=P(X\le x)=\int_{-\infty}^{1} 0\,du+\int_{1}^{2}(u-1)\,du+\int_{2}^{3}(3-u)\,du+\int_{3}^{x}0\;du$

$$=\int_{-\infty}^{1} 0\,du+\int_{1}^{2}(u-1)\,du+\int_{2}^{3}(3-u)\,du+\int_{3}^{x}0\,du$$

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